PROJECT DEVELOPMENT PHASE
Data Preprocessing Steps, Business Questions and Visualization

Date	01 JULY 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	4 Marks

Data Preparation Steps

Step 1: Project Initialization

- Created the Personalized Networking Assistant project structure.
- Configured the FastAPI backend and Streamlit frontend.
- Verified that all project folders and Python files were created successfully.
- Prepared the development environment for implementation.

Step 2: Install Required Libraries

Installed all required Python packages.

- FastAPI
- Uvicorn
- Streamlit
- Requests
- Wikipedia-API
- JSON
- Python Standard Libraries

Verified that all libraries were installed successfully.

Step 3: Configure Backend Services

Developed backend services including:

- Event Analyzer Service
- Conversation Generator Service
- Knowledge Assistant Service

- History Logger Service
- Feedback Logger Service

Verified that all API endpoints were functioning correctly.

Step 4: Configure Frontend

Developed an interactive Streamlit web interface.

Features included:

- Event Description
- Knowledge Assistant
- Save Networking Session
- Feedback Form
- Project Summary
- Notifications
- Sidebar Navigation

Step 5: API Integration

Integrated the Streamlit frontend with the FastAPI backend using REST APIs.

Verified communication between:

- Streamlit
- FastAPI
- Wikipedia API

Step 6: Local Data Storage

Configured JSON files for local storage.

Created:

- history. json
- feedback. json

Verified successful read/write operations.

Step 7: Final Validation

Verified that:

- Event analysis worked correctly.
- Conversation starters were generated successfully.
- Wikipedia knowledge retrieval worked correctly.
- Session history was saved successfully.
- Feedback submission worked correctly.
- All modules were integrated successfully.

Business Questions with Visualization

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1	Can the application generate intelligent conversation starters from an event description?	Conversation Starter Cards
2	Can users retrieve verified knowledge from Wikipedia?	Knowledge Assistant Card
3	Can networking sessions be stored successfully?	Session Save Status
4	Can user feedback be collected successfully?	Feedback Confirmation
5	What networking goal has the user selected?	Goal Information Card
6	What profession is selected by the user?	User Profile Card
7	How many services are integrated into the application?	Project Summary
8	Which technologies are used in the application?	Technology Stack Card

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Visualizations Used

1. User Profile Section

Displays:

- User Name
- Profession
- Networking Goal

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2. Event Description Module

Displays:

- Event Input
- Generate Conversation Starters Button
- Generated Conversation Starter Cards

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3. Knowledge Assistant

Displays:

- Topic Input
- Verified Information
- Wikipedia Summary Card

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4. Session History Module

Displays:

- Save Session Button
- Success Notification

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5. Feedback Module

Displays:

- Feedback Text Area
- Submit Feedback Button
- Success Message

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6. Project Summary

Displays:

- Features
- Technologies Used
- Application Overview

API Endpoints Developed

Endpoint	Purpose
/	Home Page
/generate	Generate Conversation Starters
/factcheck	Verify Knowledge
/save-history	Save Networking Session
/feedback	Save User Feedback

Modules Implemented

Frontend

- Streamlit

Backend

- FastAPI

Services

- Event Analyzer
- Conversation Generator
- Knowledge Assistant
- History Logger
- Feedback Logger

External API

- Wikipedia API

Storage

- JSON Files

Expected Outputs

The developed application successfully provides:

- AI-generated conversation starters
- Verified knowledge using Wikipedia

- Networking session management
- Feedback collection
- Interactive Streamlit interface
- FastAPI backend services
- JSON-based data storage
- Responsive user experience

Conclusion

The **Personalized Networking Assistant (Smart Connect AI)** was successfully developed by integrating **FastAPI**, **Streamlit**, **Wikipedia API**, and **JSON storage**. The application enables users to generate AI-powered conversation starters, verify networking topics, save networking sessions, and submit feedback through an intuitive interface. The development process ensured proper API integration, modular architecture, and reliable functionality, resulting in a scalable and user-friendly networking assistant.